



Gibbs free energy minimization for the calculation of chemical and phase equilibrium using linear programming

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ABSTRACT

One important concern in the calculation of chemical and phase equilibrium using Gibbs free energy minimization is how to guarantee finding the global optimum without depending on an initial guess. This work proposes an approach for the minimization of the Gibbs free energy using linear programming that guarantees finding the global optimum within some level of precision, for any kind of thermodynamic model. The strategy was used in the calculation of chemical and phase equilibrium involving binary and ternary systems at low and high pressure. The method presented in this proposal is easy to implement, robust and can use several thermodynamic models.

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1. Introduction

Growing concerns with the environment and the regulation of tough environmental laws have stimulated the use of techniques for the improvement of more precise and efficient chemical and physical processes of separation. The calculation of chemical and phase equilibrium has an important application in solving separation problems. Due to this, there are in the literature many works that present several mathematical methodologies with this aim [1–9].

The necessary and sufficient conditions to achieve equilibrium in a multiphase multicomponent system, at constant temperature and pressure, is the global minimum of the Gibbs free energy of the system. Based on this principle, equilibrium problems may be formulated and solved as optimization problems [10–17]. The objective function for these problems is nonlinear and usually non-convex, so that methods of global optimization are essential for its resolution.

The calculation of chemical and phase equilibrium using nonlinear programming for the Gibbs free energy minimization has been used for some time [11,18,19]. For a convex thermodynamic model, the global optimum can be found more easily [20,21]. However, for a nonconvex thermodynamic model the problem is more difficult to solve, due to the existence of several local optima [4–7,10–19], so that sometimes a reliable initial guess is necessary.

The calculation of equilibrium through minimization methods can also be done using linear programming (LP), if some adequate strategy is used. White et al. [20], Bullard and Biegler [22], Gopal and Biegler [23], Han and Rangaiah [24], Zhu and Xu [25], Lin and Stadtherr [26,27] used LP in the calculation of chemical and phase equilibrium. White et al. [20] linearized the Gibbs free energy function for the hypothesis of ideal behavior for the liquid and vapor phases. Bullard and Biegler [22] carried out the calculation of vapor–liquid equilibrium (VLE) using the hypothesis of ideal and nonideal behavior (UNIQUAC) for the liquid phase employing the linear programming with iteration proposed by Bullard and Biegler [28]. Gopal and Biegler [23] extended the capacity of the linear programming with iteration proposed by Bullard and Biegler [28] for nonsmooth problems such as dynamic simulation problems involving nonideal three-phase systems. Han and Rangaiah [24] developed a method to solve systems of algebraic equations (τ -method) for the calculation of multiphase equilibrium using the successive linear programming (SLP) method. Zhu and Xu [25] developed a modified Branch-Bound algorithm to minimize the distance of the tangent plane using the hypothesis of nonideal behavior (UNIQUAC) for the liquid phase. Lin and Stadtherr [26,27] used the interval method for the calculation of chemical and phase equilibrium employing LP techniques.

The present work proposes a strategy that is able to find the global optimum of the Gibbs free energy, within some level of precision, for any kind of thermodynamic model, based on a linear programming model. The molar fraction vectors of a relatively large number of potential phases are fixed as parameters in a

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way that all quantities that depend on them (fugacity and activity) will also be considered parameters in the model. Thus, the variables of the linear programming problem are only the total number of moles of each potential phase. Typically, once the solution of the linear programming problem is found, most of the initial potential phases have zero total mass and therefore are not present at equilibrium, i.e., only a few out of all potential phases (whose composition vectors are set at the beginning of the algorithm) are stable within some prescribed level of precision set on the composition of the phases. The proposed methodology was applied in the calculation of chemical and phase equilibrium for systems known to be difficult from other studies in the literature.

2. Methodology

2.1. Minimization of Gibbs free energy

The calculation of phase equilibrium corresponds to obtaining the minimum of the Gibbs free energy of the system, at constant temperature (T) and pressure (P), with respect to the number of moles of each component in each phase (n_i^k). For a system with NP phases and NC components, we have:

$$G = \sum_{k=1}^{NP} \sum_{i=1}^{NC} n_i^k \mu_i^k \quad (1)$$

where n_i^k is the number of moles of component i in k phase, and μ_i^k is the chemical potential of component i in k phase, in which the chemical potential is a function of k phase composition, temperature (T), pressure (P). NP is the number of phases and NC is the number of components.

Eq. (1) can be rewritten in terms of the fugacities ($\hat{f}_i^k$) [18]:

$$G = \sum_{k=1}^{NP} \sum_{i=1}^{NC} n_i^k \left[\mu_i^\circ + RT \ln \frac{\hat{f}_i^k}{f_i^\circ} \right] \quad (2)$$

where μ_i° is the chemical potential of pure component i in a reference state, f_i° is the fugacity of pure component i at standard state and R is the gas universal constant. The standard state for vapor phase is taken as an ideal gas at system temperature and pressure of 1 bar, for the liquid phase is taken as the fugacity of pure component i in liquid phase at system temperature and for solid phase the reference state was the solid phase at 298.15 K and 1 bar.

The fugacities for the calculation of liquid–vapor, liquid–liquid, solid–vapor and liquid–liquid–vapor equilibrium (LLVE) at high pressures can be calculated by Eqs. (3)–(5) [30]:

$$\hat{f}_i^V = y_i \hat{\phi}_i^V P \quad (3)$$

$$\hat{f}_i^L = x_i \hat{\phi}_i^L P \quad (4)$$

$$\hat{f}_i^S = f_i^{\circ,L} \exp \left[\frac{(V_i^S - V_i^{\circ,L})(P - P_i^{sub})}{RT} + \frac{\Delta h_i^{fus}}{RT_i^{fus}} \left(1 - \frac{T_i^{fus}}{T} \right) \right] \quad (5)$$

where y_i and x_i are the molar fraction of component i in the gaseous and the liquid phase, respectively, $\hat{\phi}_i^V$ and $\hat{\phi}_i^L$ are the fugacity coefficient for component i in gaseous and liquid phase, $f_i^{\circ,L}$ is the fugacity for component i in the reference state at system temperature and reference pressure, V_i^S is the solid molar volume for component i and $V_i^{\circ,L}$ is the heavy compound molar volume in the sub-cooled liquid state for component i , P_i^{sub} is the sublimation pressure of the solid for component i , and Δh_i^{fus} is the molar fusion enthalpy at normal fusion temperature (T_i^{fus}) for pure component i .

The coefficients of fugacity ($\hat{\phi}_i^k$), for high pressure systems, were calculated using Peng–Robinson (PR) equation of state (EOS) employing the quadratic mixing rule proposed by Van der Waals (VdW2).

For low pressure systems and temperatures below the critical point, the fugacities can be calculated for ideal gas and nonideal solutions using Eqs. (6) and (7):

$$\hat{f}_i^V = y_i P \quad (6)$$

$$\hat{f}_i^L = x_i \gamma_i P_i^{sat} \quad (7)$$

where P_i^{sat} is the saturation pressure of the liquid and γ_i is the activity coefficient for pure component i .

The activity coefficients were calculated using the thermodynamic models of Wilson, NRTL and UNIQUAC [29].

As temperature and pressure are given for each system, Δh_i^{fus} , T_i^{fus} , and V_i^S can be obtained from thermodynamic data bank, P_i^{sat} , P_i^{sub} , μ_i° , $f_i^{\circ,L}$, and $V_i^{\circ,L}$ are parameters which can be previously calculated (before minimizing G). The fugacity of the sub-cooled liquid, $f_i^{\circ,L}$, and the specific volume in the sub-cooled liquid reference state, $V_i^{\circ,L}$, can be calculated using an EOS by [30]:

$$\ln \left(\frac{f_i^{\circ,L}}{P} \right) = \int_0^P \frac{Z_i - 1}{P} dP \quad (8)$$

where Z_i is the compressibility factor.

The calculation of the chemical potential for the pure component i in the reference state, μ_i° , at any temperature, can be calculated for the system conditions using the following known thermodynamic relations [31] (see Appendix A):

$$\frac{\partial}{\partial T} \left(\frac{G_i}{RT} \right)_P = - \frac{H_i}{RT^2} \quad (9)$$

$$\left(\frac{\partial H_i}{\partial T} \right)_P = C_{p,i} \quad (10)$$

where G_i and H_i are, respectively, the partial molar Gibbs free energy and partial molar enthalpy for component i and $C_{p,i}$ is the heat capacity for component i .

The saturation pressures for the liquid, P_i^{sat} , and sublimation pressures for the solid, P_i^{sub} , were calculated using Antoine equation:

$$\ln P_i^{sat} = A_i^{sat} - \frac{B_i^{sat}}{C_i^{sat} + T} \quad (11)$$

$$\ln P_i^{sub} = A_i^{sub} - \frac{B_i^{sub}}{C_i^{sub} + T} \quad (12)$$

where A_i , B_i and C_i are Antoine parameters for each component i .

In the Gibbs free energy minimization (Eq. (1)), the restrictions of mass balance and non-negativity of number of moles must be observed, according to the following equations:

$$n_i^k \geq 0, \quad i = 1, \dots, NC, \quad k = 1, \dots, NP \quad (13)$$

- for non-reacting components:

$$\sum_{k=1}^{NP} n_i^k = n_i^0, \quad i = 1, \dots, NC \quad (14)$$

- for reacting components:

$$\sum_{k=1}^{NP} \sum_{i=1}^{NC} a_{mi} n_i^k = \sum_{i=1}^{NC} a_{mi} n_i^0, \quad m = 1, \dots, NE \quad (15)$$

where a_{mi} is the number of atoms of element m in component i , n_i^0 is the initial number of moles of component i and NE is the number of elements in the system.

2.2. Proposed strategy

The first step in the method is the definition of a grid in the mole fraction domain. The number of dimensions of this grid equals the number of components of the multicomponent system. Every point of the grid that matches the constraint of a unity mole fraction sum corresponds to the composition vector of one or more phases that could potentially exist at equilibrium. The minimum number of potential phases equals the number of composition vectors. The finer is the grid the greater is the number of potential phases.

In order to explain the proposed strategy used in this paper, consider a mixture with any number of phases and 3 components ($NC=3$). For each phase k , the domain of compositions vectors (z_1^k, z_2^k, z_3^k) is mapped and discretized in a number of points. Each interval $0 \leq z_i^k \leq 1$ is divided in N intervals with equal length δ , such that:

$$\delta = \frac{1}{N} \quad (16)$$

Since the sum of the molar fractions in each phase must be equal to 1, the total number of points generated by this procedure, M , for three components, is given by:

$$M = \frac{(N+1)(N+2)}{2} \quad (17)$$

and the molar fractions for each point is given by (see Appendix B):

$$z_1^k = 1 - \delta(p-1), \quad k = 1, \dots, M \quad (18)$$

$$z_2^k = \delta \left[\frac{p(p+1)}{2} - k \right], \quad k = 1, \dots, M \quad (19)$$

$$z_3^k = \delta \left[k - 1 - \frac{p(p-1)}{2} \right], \quad k = 1, \dots, M \quad (20)$$

where

$$p = \text{floor} \left[\frac{\sqrt{8k-7}-1}{2} \right] + 1, \quad k = 1, \dots, M \quad (21)$$

or

$$p = \text{ceil} \left[\frac{\sqrt{8k+1}-1}{2} \right], \quad k = 1, \dots, M \quad (22)$$

and where floor and ceil are two mathematical functions used for rounding a real number to its closest integer values (for example: floor(1.99)=1 and ceil(1.99)=2; floor(1)=1 and ceil(1)=1).

Each one of these points is then considered as a potential phase k , with a given composition vector (z_1^k, z_2^k, z_3^k). In this way, all quantities that depend only on composition, and T and P , will be parameters in the minimization of G , so that μ_i^k is also a parameter. The resulting problem is then a linear programming in the variables n_i^k , which are the number of moles of component i in the potential phase k , where the previously established phase composition vectors give the molar fraction of each component at each potential phase k . Therefore, the variables n_i^k should satisfy the linear restrictions:

$$n_i^k = z_i^k \sum_{j=1}^{NC} n_j^k, \quad i = 1, \dots, NC, \quad k = 1, \dots, M \quad (23)$$

This approach can then be applied in two different ways, depending on how the thermodynamic model is formulated:

2.2.1. Gamma-phi approach

For liquid–vapor equilibrium (LVE), the nonideality for the liquid phase is given by the activity coefficients, γ_i , while the nonideality for the vapor phase is given by the fugacity coefficients, $\hat{\phi}_i$, using different thermodynamic models for the two phases. In this work, it was considered that $\hat{\phi}_i = 1$.

For this approach, the composition vectors x_i^k and y_i^k are given by Eqs. (18)–(20), so that they are parameters in the minimization of G . There are M liquid potential phases and M vapor potential phases. Therefore, in this approach the number of potential phases is $NP=2M$. The number of moles for each component i , for each potential phase k in the liquid phase and for each potential phase k in the vapor phase, is given by:

$$n_i^{L,k} = x_i^k \sum_{j=1}^{NC} n_j^{L,k}, \quad i = 1, \dots, NC, \quad k = 1, \dots, M \quad (24)$$

$$n_i^{V,k} = y_i^k \sum_{j=1}^{NC} n_j^{V,k}, \quad i = 1, \dots, NC, \quad k = 1, \dots, M \quad (25)$$

The independent variables in the minimization are then the mole number for each component i , for each potential liquid phase, $n_i^{L,k}$, and each potential vapor phase, $n_i^{V,k}$, since all others are calculated from them.

2.2.2. Phi-phi approach

For liquid–vapor, liquid–liquid and liquid–liquid–vapor equilibrium, the nonideality of all phases are given by the fugacity coefficients, $\hat{\phi}_i^k$, using the same equation of state for all phases.

For LLE, the number of potential phases is $NP=M$, with composition vectors given by Eqs. (18)–(20), since all real phases have different compositions.

For LVE and LLVE problems without homogeneous azeotropes, it is enough to consider $NP=M$, with composition vectors z_i^k given by Eqs. (18)–(20), since all real liquid and vapor phases have different compositions.

However, if homogenous azeotropes occur at some LVE problem, then potential phases with different densities and with same compositions should also be considered. Two composition vectors $z_i^{\alpha,k}$ and $z_i^{\beta,k}$ are then defined, with values given by Eqs. (18)–(20), without previously specifying which phase (α or β) would be liquid or vapor, so that they are parameters in the minimization of G . Therefore, the number of potential phases is $NP=2M$. The number of moles for each component i , for each potential phase k , is given by:

$$n_i^{\alpha,k} = z_i^{\alpha,k} \sum_{j=1}^{NC} n_j^{\alpha,k}, \quad i = 1, \dots, NC, \quad k = 1, \dots, M \quad (26)$$

$$n_i^{\beta,k} = z_i^{\beta,k} \sum_{j=1}^{NC} n_j^{\beta,k}, \quad i = 1, \dots, NC, \quad k = 1, \dots, M \quad (27)$$

The independent variables in the minimization are then the number of moles for each component i at each potential phase, $n_i^{\alpha,k}$ and $n_i^{\beta,k}$, since all others are calculated from them. After minimization of G , subject to all constraints in the variables $n_i^{\alpha,k}$ and $n_i^{\beta,k}$, the identification of which phase is vapor or liquid is done by the density of the phase, which depends on the composition ($z_1^{\alpha,k}, z_2^{\alpha,k}, z_3^{\alpha,k}$), ($z_1^{\beta,k}, z_2^{\beta,k}, z_3^{\beta,k}$) and T and P .

Table 1
NRTL parameters obtained from McDonald and Floudas [2].

Components	$g_{ij}-g_{ji}$		$g_{ji}-g_{ii}$		$\alpha_{ij} = \alpha_{ji}$	
	300 K	333 K	300 K	333 K	300 K	333 K
Benzene–acetonitrile	693.61	998.2	92.47	65.74	0.67094	0.88577
Benzene–water	3892.44	3883.2	3952.2	3849.57	0.23906	0.24698
Acetonitrile–water	415.38	363.57	1016.28	1262.4	0.20202	0.3565

$g_{ij}-g_{ji}$, binary interaction parameters for the NRTL model (cal/mol); $\alpha_{ij} = \alpha_{ji}$, binary interaction parameters for the NRTL model (dimensionless).

Table 2
Calculated number of moles: benzene (1) + acetonitrile (2) + water (3).

Δx (%)		Liquid I			Liquid II			Vapor		
		1	2	3	1	2	3	1	2	3
Case 1 ($T = 333$ K and $P = 0.769$ atm)										
lit ^a		0.2395	0.2270	0.0337	0.0007	0.0217	0.2624	0.1046	0.0616	0.0524
$\delta_1 = 0.0100$	0.65	0.1022	0.1066	0.0198	0.0000	0.0162	0.2636	0.1273	0.0786	0.0650
$\delta_2 = 0.0050$	0.63	0.2134	0.2089	0.0318	0.0014	0.0234	0.2509	0.1300	0.0780	0.0657
$\delta_3 = 0.0025$	0.41	0.2181	0.2135	0.0325	0.0007	0.0213	0.2523	0.1260	0.0756	0.0637
$\delta_4 = 0.0020$	0.54	0.2137	0.2118	0.0330	0.0005	0.0206	0.2496	0.1306	0.0779	0.0659
$\delta_5 = 0.0016$	0.46	0.2157	0.2120	0.0324	0.0009	0.0214	0.2513	0.1282	0.0769	0.0648
Case 2 ($T = 333$ K and $P = 1$ atm)										
lit ^a		0.3440	0.2865	0.0379	0.0008	0.0238	0.3106	–	–	–
$\delta_1 = 0.0050$	0.28	0.3432	0.2849	0.0366	0.0017	0.0254	0.3119	–	–	–
$\delta_2 = 0.0025$	0.06	0.3440	0.2869	0.0385	0.0008	0.0234	0.3099	–	–	–
Case 3 ($T = 300$ K and $P = 0.1$ atm)										
lit ^a		3×10^{-6}	5×10^{-4}	0.0159	–	–	–	0.3448	0.3099	0.3326
$\delta_1 = 0.0050$	0.05	0.0000	6×10^{-4}	0.0178	–	–	–	0.3448	0.3098	0.3306
$\delta_2 = 0.0025$	0.18	0.0000	4×10^{-4}	0.0149	–	–	–	0.3448	0.3097	0.3307

^a McDonald and Floudas [2].

3. Results and discussion

In order to evaluate the proposed methodology, some case studies from literature were selected involving systems in thermodynamic equilibrium at low and high pressures, requiring different levels of mathematical and computational difficulty in finding the solution.

The proposed methodology was implemented in GAMS® 2.5 (“General Algebraic Modeling System”), using the CPLEX solver, and executed in Pentium III (512 MB, 900 MHz).

The thermodynamic data used were obtained from Reid et al. [32,33] and Poling et al. [29], DIADEM [34] and the National Institute of Standards and Technology [35].

In order to compare results found from the literature and the results calculated in this work, the average deviation of molar fractions was calculated:

$$\Delta x = 100 \sqrt{\frac{\sum_k^{\text{NP}} \sum_i^{\text{NC}} [(x_{ik}^{\text{lit}} - x_{ik}^{\text{work}})^2]}{\text{NP} \cdot \text{NC}}} \quad (28)$$

where the superscript *lit* refers to literature molar fraction and *work* refers to the molar fraction obtained by the proposed approach.

3.1. Low pressure problems

3.1.1. Problem 1: benzene + acetonitrile + water

The benzene (1) + acetonitrile (2) + water (3) system in VLE with a second liquid potential phase was studied by Castillo and Grossmann [18] and McDonald and Floudas [2]. The NRTL model was used for the calculation of the activity coefficient. The parameters were obtained from McDonald and Floudas [2] and are presented in Table 1.

Table 2 shows the results obtained by the proposed methodology and the results obtained by McDonald and Floudas [2], for three conditions of temperature and pressure, for the system with

an initial composition of 0.34483 mol of benzene, 0.31034 mol of acetonitrile and 0.3483 mol of water. In Table 3 the CPU times for different intervals and the calculated values of *G* are compared.

Table 3
Comparison of results for the benzene (1) + acetonitrile (2) + water (3) system.

	δ	CPU time (s)	Δx (%)	<i>G</i> (cal)
Case 1	0.0100	0.1	0.65	–713.816
	0.0050	11.0	0.63	–714.134
	0.0025	380.5	0.41	–714.261
	0.0020	945.3	0.54	–714.249
	0.0016	2091.0	0.46	–714.253
Case 2	0.0050	12.5	0.28	–713.268
	0.0025	404.5	0.06	–7123.460
Case 3	0.0050	16.4	0.05	–2511.432
	0.0025	373.9	0.18	–2511.435

Table 4
Binary parameters – UNIQUAC model [39].

Components	$u_{ij}-u_{ji}$	$u_{ij}-u_{jj}$
SBA–DSBE	–193.140	415.850
SBA–water	424.025	103.810
DSBE–water	315.312	3922.500

$u_{ij}-u_{ji}$, binary interaction parameters for UNIQUAC (cal/mol).

Table 5
Pure component UNIQUAC parameters [39].

Components	q_i	q'_i	r_i
SBA	3.6640	4.0643	3.9235
DSBE	5.1680	5.7409	6.0909
Water	1.4000	1.6741	0.9200

q_i , q'_i and r_i dimensionless.

Table 6

Molar fractions obtained for the SBA (1) + DSBE (2) + water (3) system.

Δx (%)		Liquid I			Liquid II			Vapor		
		1	2	3	1	2	3	1	2	3
Tray 28 ($T = 363.2$ K and $P = 1.170$ atm; $Z_1 = 40.30707$, $Z_2 = 5.14979$, $Z_3 = 54.54314$)										
lit ^a		0.51802	0.05110	0.43088	0.05667	0.00000	0.94333	0.34024	0.08762	0.57214
$\delta_1 = 0.0100$	0.14	0.52000	0.05000	0.43000	0.06000	0.00000	0.94000	0.34000	0.09000	0.57000
$\delta_2 = 0.0050$	0.15	0.51500	0.05000	0.43500	0.05500	0.00000	0.94500	0.34000	0.09000	0.57000
$\delta_3 = 0.0020$	0.04	0.51800	0.05200	0.43000	0.05600	0.00000	0.94400	0.34000	0.08800	0.57200
Tray 25 ($T = 362.35$ K and $P = 1.166$ atm; $Z_1 = 35.18411$, $Z_2 = 12.55338$, $Z_3 = 52.26247$)										
lit ^a		0.52037	0.15429	0.32534	0.04401	0.00000	0.95599	0.30236	0.13931	0.55833
$\delta_1 = 0.0100$	0.21	0.52000	0.15000	0.33000	0.04000	0.00000	0.96000	0.30000	0.14000	0.56000
$\delta_2 = 0.0050$	0.08	0.52000	0.15500	0.32500	0.04500	0.00000	0.95500	0.30000	0.14000	0.56000
$\delta_3 = 0.0020$	0.09	0.5200	0.1520	0.3280	0.0440	0.0000	0.9560	0.3040	0.1380	0.5580
Tray 7 ($T = 361.67$ K and $P = 1.145$ atm; $Z_1 = 33.45195$, $Z_2 = 18.21254$, $Z_3 = 48.33661$)										
lit ^a		0.48182	0.24579	0.27239	0.03759	0.000	0.96241	0.28013	0.16429	0.55558
$\delta_1 = 0.0100$	0.48	0.4900	0.2300	0.2800	0.04000	0.000	0.96000	0.28000	0.16000	0.56000
$\delta_2 = 0.0050$	0.18	0.48500	0.24000	0.27500	0.04000	0.000	0.96000	0.28000	0.16500	0.55500
$\delta_3 = 0.0020$	0.23	0.48600	0.23800	0.27600	0.03800	0.000	0.96200	0.28200	0.16200	0.55600
$\delta_4 = 0.0016$	0.26	0.48640	0.23680	0.27680	0.03840	0.000	0.96160	0.28160	0.16320	0.55520

^a McDonald and Floudas [38].

From the average deviations in molar fractions (Δx), presented in Tables 2 and 3, it is possible to conclude that the proposed methodology agrees with the results found by McDonald and Floudas [2] for all conditions. The results did not change significantly with the interval δ , except for the case with $\delta = 0.01$. For the different values of intervals tested, it was chosen the one with lowest G . For example, for case 1, for $\delta = 0.01$ the value of $G = -713.816$ cal; for $\delta = 0.005$ the value of $G = -714.134$ cal; for $\delta = 0.0025$ the value of $G = -714.261$ cal; for $\delta = 0.002$ the value of $G = -714.249$ cal; and for $\delta = 0.0016$ the value of $G = -714.253$ cal. Therefore, for case 1 the best value is found when $\delta = 0.0025$.

3.1.2. Problem 2: SBA + DSBE + water

The system involving the dehydration of sec-butanol (SBA) (1), resulting in di-sec-butyl-ether (DSBE) (2) and water (3), presents azeotropic behavior in LLVE. Therefore, this is a system with combined chemical and phase equilibrium.

The experimental data for this system were obtained by Kovach and Seider [36] and Widagdo et al. [37]. McDonald and Floudas [38] modeled this system using the UNIQUAC model. The proposed methodology was applied to this system using the parameters obtained from McDonald and Floudas [39], presented in Tables 4 and 5. The Antoine parameters were the ones supplied by Kovach and Seider [36].

In Table 6 the results obtained by the proposed methodology and those calculated by McDonald and Floudas [38] are compared.

In Table 7 the CPU times for different intervals and the calculated values of G are compared. It can be seen that an interval of $\delta = 0.005$ is adequate. A larger interval results in less precise results, and a smaller interval results in a computational time that is too long.

Table 7

Comparison of results for the SBA (1) + DSBE (2) + water (3) system.

Tray	δ	CPU time (s)	Δx (%)	G (cal)
28	0.0100	1.3	0.14	-4,396,890.3
	0.0050	28.1	0.15	-4,396,891.9
	0.0020	1646.0	0.04	-4,396,891.8
25	0.0100	1.4	0.21	-4,234,902.6
	0.0050	28.2	0.08	-4,234,903.9
	0.0020	2119.8	0.09	-4,234,904.3
7	0.0100	1.2	0.48	-440,136.7
	0.0050	27.7	0.18	-440,136.9
	0.0020	1291.8	0.23	-440,137.1
	0.0016	5220.3	0.26	-440,137.1

Due to the sensitivity of the data for this example, it represents a challenge for the calculation of phase equilibrium [38]. Tray 7 particularly showed more difficulty in the calculation, since depending on the initial composition some of the phases formed in very small quantities, so that very small changes could make one of the phases disappear. Therefore, the initial composition used for tray 7 was 0.71062 mol of SBA, 1.31175 mol of DSBE and 6.05036 mol of water. The final results are presented in Tables 5 and 6, which are in agreement with those of McDonald and Floudas [38].

3.1.3. Problem 3: water + ethanol + hexane

Gomis et al. [40] experimentally measured the water (1) + ethanol (2) + hexane (3) ternary system in VLE. The proposed methodology was applied to the system using the set of binary interaction parameters for NRTL and UNIQUAC supplied by Gomis et al. [40] and the parameters set obtained from Gmehling and Onken [41] from DECHEMA series. In Tables 8 and 9 the binary interaction parameters set for NRTL and UNIQUAC are presented. In Table 10 the Antoine parameters obtained from Gomis et al. [40] are presented. The parameters of the pure components for UNIQUAC were obtained from DECHEMA series.

Table 8

NRTL and UNIQUAC parameters obtained by Gomis et al. [40].

Components	NRTL parameters			UNIQUAC parameters	
	$g_{ij} - g_{ji}$	$g_{ji} - g_{ii}$	$\alpha_{ij} = \alpha_{ji}$	$u_{ij} - u_{jj}$	$u_{ij} - u_{ii}$
Water–ethanol	765.2826	-889.5103	0.3031	-821.2323	-347.0423
Water–hexane	2130.053	1661.666	0.2000	1057.531	1823.642
Ethanol–hexane	421.547	275.1408	0.3827	-199.0427	564.5045

$g_{ij} - g_{ji}$, binary interaction parameters for the NRTL model (cal/mol); $u_{ij} - u_{jj}$, binary interaction parameters for UNIQUAC (cal/mol); $\alpha_{ij} = \alpha_{ji}$, binary interaction parameters for the NRTL model (dimensionless).

Table 9

NRTL and UNIQUAC parameters obtained from DECHEMA series.

Components	NRTL parameters			UNIQUAC parameters	
	$g_{12} - g_{22}$	$g_{21} - g_{11}$	$\alpha_{12} = \alpha_{21}$	$u_{ij} - u_{jj}$	$u_{ij} - u_{ii}$
Water–ethanol	1376.3536	-114.8438	0.2983	2142.6513	-375.1341
Water–hexane	3407.1000	1662.0000	0.20000	598.6900	1161.7000
Ethanol–hexane	979.419	1505.6508	0.4766	-173.3145	1174.9142

$g_{ij} - g_{ji}$, binary interaction parameters for the NRTL model (cal/mol); $u_{ij} - u_{jj}$, binary interaction parameters for UNIQUAC (cal/mol); $\alpha_{ij} = \alpha_{ji}$, binary interaction parameters for the NRTL model (dimensionless).

Table 10
Antoine equation parameters for pure substances [40].

Components	A	B	C	Temperature range (°C)
Water	8.07131	1730.630	233.426	+1/+100
Ethanol	8.11220	1592.864	226.184	+20/+93
Hexane	6.91058	1189.64	226.280	−30/+170

P_i^{sat} in mmHg and T in °C.

In Figs. 1 and 3 the comparisons of the phase equilibrium behavior among the results experimentally measured by Gomis et al. [40] and the ones calculated by the proposed methodology are presented using the binary interaction parameters from Gomis et al. [40] and DECHEMA series, respectively. The discretization interval used was $\delta = 0.005$. The calculation was done using the same initial conditions (temperature, pressure and feed composition) as reported by Gomis et al. [40]. The average time of CPU was 9.7 s and 17.2 s to NRTL and UNIQUAC model, respectively.

In Fig. 1 it is possible to observe that the results obtained using the binary interaction parameters from Gomis et al. [40], mainly for the NRTL model, did not represent adequately the experimental results. Besides, at temperature of 342.64 K, the number of phases did not coincide with the experimental results. For the UNIQUAC model with binary parameters given by Gomis et al. [40], at temperatures of 342.64 K, 341.71 K, 338.04 K, 335.28 K and 332.55 K, the calculated number of phases did not coincide with the experimental ones. However, using the binary parameters given by DECHEMA for NRTL and UNIQUAC models, the calculated number of phases was in agreement with the experimental data, as can be observed in Fig. 2.

3.1.4. Problem 4: toluene + water

The calculation of phase equilibrium for the toluene (1) + water (2) binary system in LLE was carried out by Castillo and Grossmann [18] and McDonald and Floudas [2]. For the simulation of this system McDonald and Floudas [2] employed the MINOS5.1 solver and the GOP algorithm. The MINOS5.1 solver obtained only the local solution while the GOP algorithm found the global solution. The binary interaction parameters for temperature of 298 K for the NRTL $\tau_{1,2} = 4.93$; $\tau_{2,1} = 7.77$ and $\alpha_{1,2} = \alpha_{2,1} = 0.2485$ model, both dimensionless, were supplied from McDonald and Floudas [2] and Bender and Block [42].

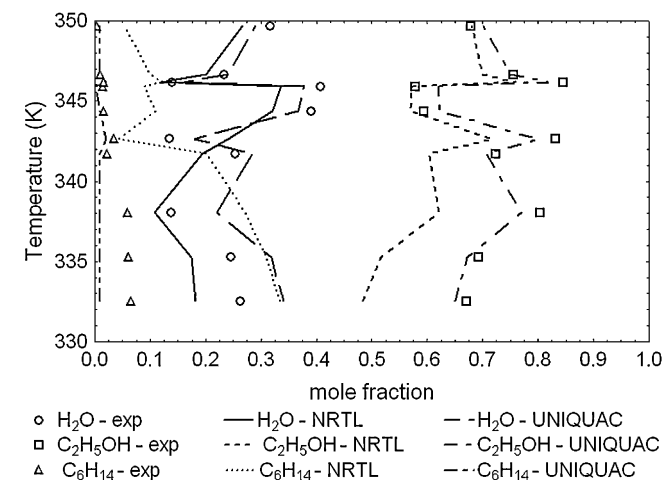


Fig. 1. ELV water (1) + ethanol (2) + hexane (3) system at 1 atm. Comparing molar fractions from experimental data by Gomis et al. [40] and calculated values using the proposed approach with NRTL and UNIQUAC models. Binary interaction parameters came from Gomis et al. [40].

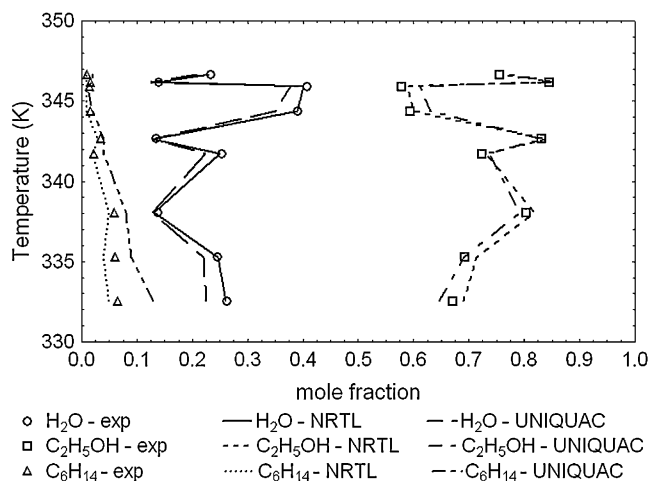


Fig. 2. ELV water (1) + ethanol (2) + hexane (3) system at 1 atm. Comparing molar fractions from experimental data by Gomis et al. [40] and calculated values using the proposed approach with NRTL and UNIQUAC models. Binary interaction parameters came from DECHEMA series.

In Table 11 the results obtained by the proposed methodology for different intervals and the ones by McDonald and Floudas [2] are presented. In Table 12 the CPU times and the calculated values of G are compared for different intervals. The molar fraction deviation between the calculated values and the ones obtained by McDonald and Floudas [2] demonstrates that the results of the proposed methodology were also able to find the global optimum. It can be seen that an interval of $\delta = 0.0001$ is adequate. A large interval results in less precise results, and a smaller interval results in a computational time that is too long.

3.2. High pressure problems

3.2.1. Problem 5: naphthalene + CO₂

The naphthalene (1) + CO₂ (2) system was widely experimentally investigated and modeled employing several techniques of deterministic and heuristic optimization to predict its phase behavior [43–50]. The proposed methodology was used to predict the solid–fluid equilibrium (SFE) using the EOS PR with the binary interaction parameters $k_{12} = 0.079175$ and $l_{12} = -0.036082$, $\Delta h_1^{fus} = 19318.40$ J/mol, $T_1^{fus} = 353.45$ K and $V_{1,0}^{S} = 111.94$ cm³/mol obtained by Corazza et al. [50].

Table 11
Molar fractions obtained for the toluene (1) + water (2) system at 298 K, 1 atm and equimolar feed.

Components	Δx (%)	Liquid I		Liquid II	
		1	2	1	2
lit ^a		0.99754	0.00255	0.00010	0.99990
$\delta_1 = 0.00500$	0.1700	0.99500	0.00500	0.00000	1.00000
$\delta_2 = 0.00100$	0.0300	0.99700	0.00300	0.00000	1.00000
$\delta_3 = 0.00010$	0.0033	0.99745	0.00260	0.00010	0.99990
$\delta_4 = 0.00001$	0.0003	0.99745	0.00255	0.00010	0.99990

^a McDonald and Floudas [2].

Table 12
Comparison of results for the toluene (1) + water (2) system.

δ	CPU time (s)	Δx (%)	G (cal)
0.00500	>0.1	0.1700	−14,758.651
0.00100	>0.1	0.0300	−14,758.885
0.00010	2	0.0033	−14,758.925
0.00001	452	0.0003	−14,758.925

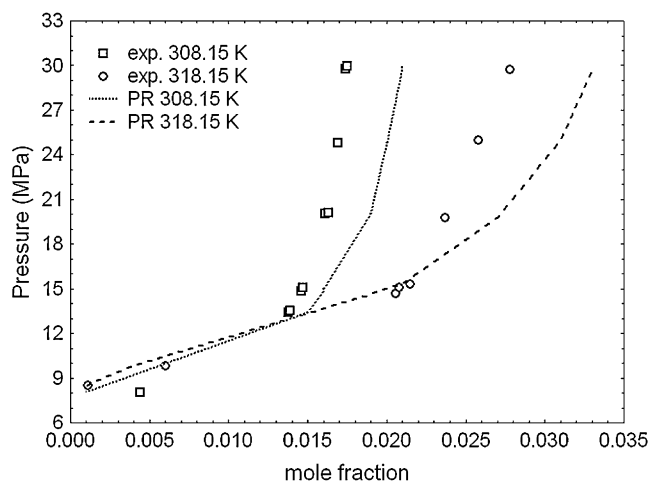


Fig. 3. Comparison between experimental solubility by Sauceau et al. [48] and calculated solubility using EOS PR of naphthalene in supercritical CO₂ at 308.15 K and 318.15 K.

Fig. 3 shows a comparison between experimental data measured by Sauceau et al. [48] and results calculated by the proposed methodology, with $\delta=0.001$. The calculation was done using the same initial conditions (temperature, pressure and feed composition) reported by Sauceau et al. [48]. The solid phase contains only naphthalene, so no discretization was needed for this phase. The average CPU time required in each calculation was 7.3 s and 7.2 s for temperatures 308.15 K and 318.18 K, respectively.

In Fig. 3 it is possible to observe that the results obtained are in agreement with experimental data and results calculated by other authors [43–50] using other methodologies.

3.2.2. Problem 6: carbon dioxide + trans-2-hexen-1-ol

The carbon dioxide (1) and trans-2-hexen-1-ol (2) system has known difficulties in calculating the global optimum [4,51]. The P - x - y diagram is presented in Fig. 4. This system was modeled employing the proposed methodology (with $\delta=0.0001$) using EOS PR and the experimental data obtained by Stradi et al. [4]. The binary interaction parameters used, $k_{12}=0.084$ and $l_{12}=0$, were obtained from Stradi et al. [4]. Fig. 4 was constructed using the experimental points from Stradi et al. [4], and the complete diagram was calculated using different values of pressure and feed composition, at constant temperature.

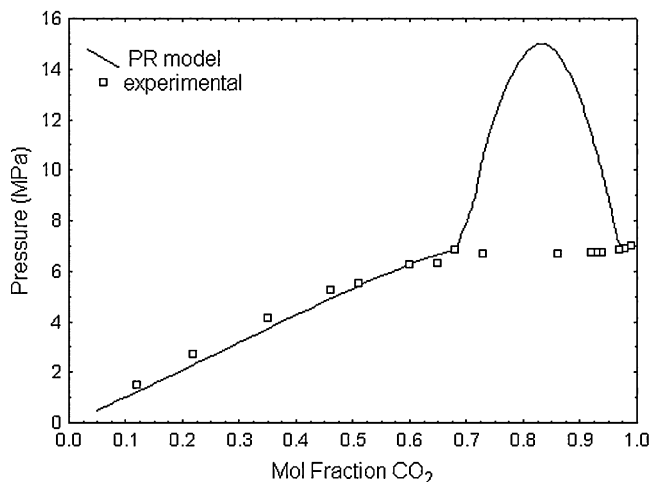


Fig. 4. P - x - y diagram for CO₂ (1) + trans-2-hexen-1-ol (2) at 303.15 K. The points indicate experimental measurements by Stradi et al. [4] and the lines indicate predictions from the EOS PR.

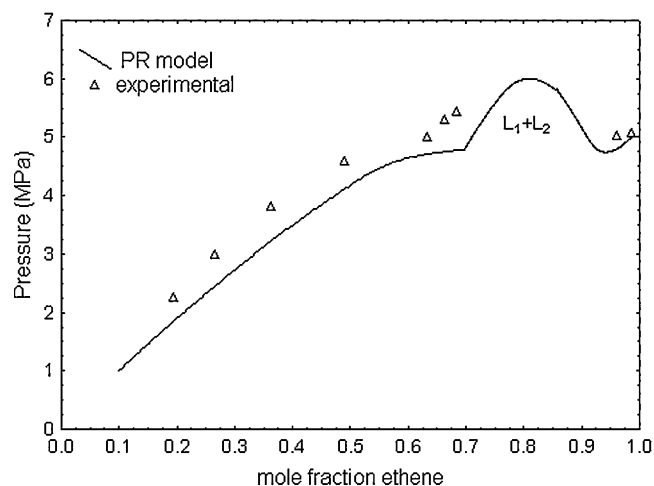


Fig. 5. P - x - y diagram for the ethene (1) + 1-propanol (2) system at high pressure and temperature of 283.65 K. Comparison between experimental data by Kodama et al. [52] and calculated results using EOS PR.

The numerical results obtained by the proposed methodology are in agreement with those obtained by Stradi et al. [4], although not in agreement with the experimental data. The CPU time was 6.9 s, on average.

3.2.3. Problem 7: ethene + 1-propanol

Kodama et al. [52] experimentally measured the ethene (1) + 1-propanol (2) system for temperature of 283.65 K and from 2.2 MPa to 5.4 MPa using EOS SRK with mixture rule of VdW2 in order to model the system. The values between the compositions calculated by Kodama et al. [52] and experimental data compositions presented significant differences.

Fig. 5 presents the experimental data for the ethene (1) + 1-propanol (2) system and the values calculated using the proposed methodology (with $\delta=0.0005$) via EOS PR employing Van der Waals mixing rule with binary interaction parameters $k_{12}=0.0092$ obtained by Freitag et al. [53]. Fig. 4 was constructed using experimental points from Ref. [52], and the complete diagram was calculated using different values of pressure and feed composition, at constant temperature. The results obtained are in agreement with those experimentally measured by Kodama et al. [52], which also shown the formation region of liquid–liquid equilibrium evident. The CPU time was 8.3 s, on average.

3.2.4. Problem 8: ethene + water + 1-propanol

Freitag et al. [54] experimentally measured the ethene (1) + water (2) + 1-propanol (3) ternary system. Subsequently Freitag et al. [53] employed several mixture rules using EOS PR to model the ternary systems, which involve binary interaction parameters obtained from binary and ternary mixtures.

In Fig. 6 the values experimentally measured by Freitag et al. [54] at 293.15 K and 80.8 bar with initial composition of 0.5 mol of ethene, 0.3 mol of water and 0.2 mol of 1-propanol are presented, and the results calculated by the proposed methodology (with $\delta=0.0005$) using two sets of binary interaction parameters are found in Table 13.

The sets of binary interaction parameters ‘a’ and ‘b’ were calculated by Freitag et al. [53]. The set ‘a’ was calculated only from experimental data of binary mixtures and the set ‘b’ was obtained from experimental data of ternary mixtures.

The calculation of LLV equilibrium is shown in Fig. 6 for a pressure of 80.8 bar, temperature of 293 K and initial composition of 0.5 mol for ethene, 0.3 mol for water and 0.2 mol for 1-propanol. The value for the Gibbs free energy for this system (after minimization)

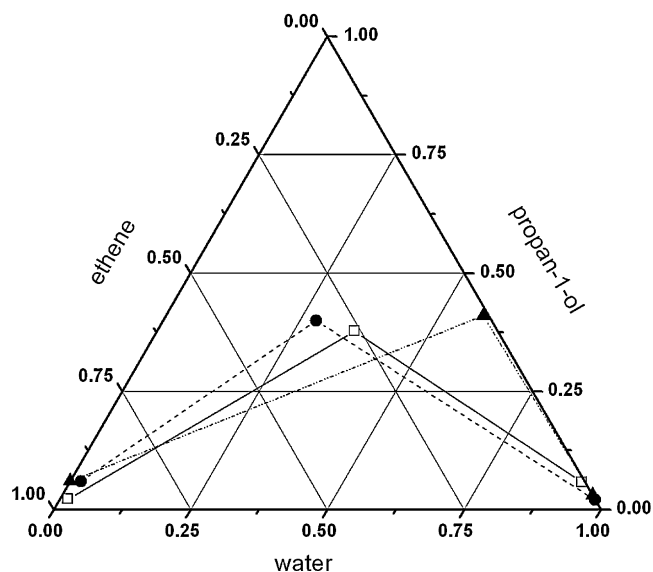


Fig. 6. L_1L_2V equilibrium for the ethene (1) + water (2) + 1-propanol (3) ternary system at 293 K and 80.8 bar. Comparison between molar fractions from experimental results by Freitag et al. [54] (□), and the results calculated by the proposed methodology using EOS PR with parameters set 'a' (Δ) and parameters set 'b' (●).

was -16215.826 cal using parameter 'a' and -16253.631 cal using parameter 'b'. The CPU times were 54.7 s and 52.5 s for parameters 'a' and 'b', respectively. It is possible to observe in Fig. 6 that the results calculated using the binary interaction parameters set 'b' are closer to the experimental results than the results calculated using the binary interaction parameters set 'a'.

In Fig. 7 the phase diagrams containing the values calculated by the proposed methodology using the parameters set 'a' and 'b' and the experimental data measured by Freitag et al. [54] at 293.15 K, 313.15 K, and 333.15 K are presented. The calculation was done using the same initial conditions (temperature, pressure and feed composition) presented by Freitag et al. [54]. The calculated results corroborate the results presented in Fig. 6. The formation of ethene

at L_2 phase is not observed for any temperature when the parameters set 'a' is used. However when employing the parameters set 'b', the formation of ethene at L_2 phase and the agreement with experimental values are noticed. The CPU times were 56.1 s for the parameter 'a' and 53.4 s for the parameter 'b', on average.

4. Conclusions

This work theoretically guarantees finding the global minimum of Gibbs free energy, within some prescribed level of precision for composition of the phases at equilibrium, using a strategy based on a discretization of the molar fraction domain. The method regards each set of molar fraction values, i.e., each composition vector, as the composition of a potential equilibrium phase. Thus, each composition vector becomes a set of scalar parameters in the minimization problem, which is established as a linear programming problem whose optimization variables are the number of moles of the phases.

The average percent deviations for the phase compositions at equilibrium, with respect to optima previously reported in the literature, were satisfactory in this work using the proposed methodology for the calculation of simultaneous phase and chemical equilibrium involving binary and ternary systems in VLE, LLE, VLE and SFE for known complex mixtures. The proposed methodology did not present any restrictions in relation to the type of thermodynamic models used, which made the use of other thermodynamic models possible. A priori any thermodynamic model can be used with the proposed methodology.

The proposed methodology was also tested for systems with four or more components. In this case, the number of variables increases considerably making the computational effort excessive for a short calculation time and also memory requirements, making the completion of the calculations not feasible. The same occurs even when using a lower level of discretization. However, the GAMS software was shown to be friendly in the implementation of the proposed methodology.

Finally, the existence of a limit for the degree of discretization in some cases, such as asymmetric and polymeric systems, may limit the applicability of the proposed technology. Nevertheless, it is possible to avoid this limitation, using the results found by the proposed approach as an initial estimate for a more refined calculation using other approaches, such as the conditions of isofugacity. Since the proposed approach can find reliable values for the number of phases and their composition, these results can be used as initial estimate when solving the nonlinear system of isofugacity equations with the aim of obtaining more accurate compositions of the equilibrium phases.

List of symbols

a_{mi}	number of atoms of element m in component i
A_i	parameter in Antoine equation for component i
B_i	parameter in Antoine equation for component i
C_i	parameter in Antoine equation for component i
$C_{p,i}$	heat capacity for component i
$\hat{f}_i^k$	fugacity for component i in the mixture in phase k
$\hat{f}_i^\circ$	fugacity for pure component i in the reference state at system temperature and reference pressure
$\hat{f}_i^{\circ,L}$	fugacity for component i in the state of sub-cooled liquid
G	Gibbs free energy for the system

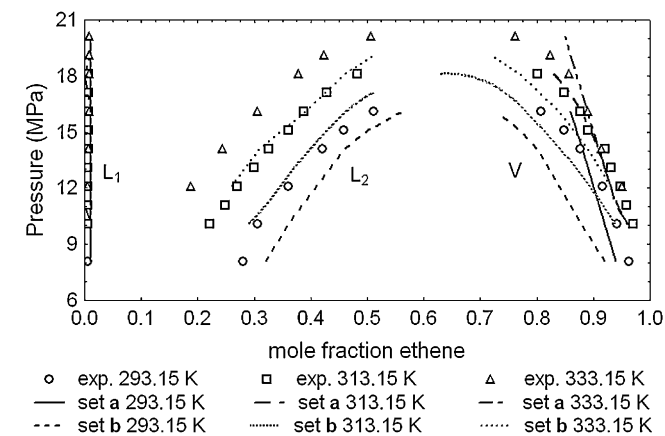


Fig. 7. L_1L_2V equilibrium for the ethene (1) + water (2) + 1-propanol (3) system at 293.15 K, 313.15 K and 333.15 K. Comparison between molar fractions of experimental data by Freitag et al. [54] and prediction of results by the proposed methodology using EOS PR with parameters sets 'a' and 'b'.

Table 13

Binary interaction parameters for the EOS PR (VdW2) for the ethane (1) + water (2) + 1-propanol (3) system supplied by Freitag et al. [53].

	k_{12}	k_{13}	k_{23}
a. Binary mixtures	0.3499	0.0092	-0.1449
b. Ternary mixtures	-0.1525	-0.0042	-0.1472

G_i	partial molar Gibbs free energy for component i
H_i	partial molar enthalpy for component i
n_i^k	number of moles for component i in phase k
n_i^0	initial number of moles for component i
NC	number of components in the system
NE	number of elements in the system
NP	number of potential phases in the system
P	absolute pressure of system
p_{sub}^i	sublimation pressure of the solid for component i
p_i^{sat}	vapor–liquid saturation pressure for component i
R	universal gas constant
T	temperature
T_i^{fus}	normal fusion temperature for component i
V_i^S	solid molar volume for component i
$V_i^{\circ,L}$	heavy compound molar volume in the sub-cooled liquid state for component i
x_i	molar fraction in liquid phase for component i
y_i	molar fraction in gas phase for component i
z_i^k	molar fraction for component i in phase k
Z_i	compressibility factor for component i

Greek symbols

μ_i^k	chemical potential for component i in phase k
μ_i°	chemical potential of reference at system temperature and reference pressure for pure component i
$\hat{\phi}_i^k$	fugacity coefficient for component i in phase k
γ_i	activity coefficient for pure component i
Δh_i^{fus}	molar fusion enthalpy for pure component i

Superscripts

V	vapor phase
L	liquid phase
S	solid phase

Subscripts

i	component in the mixture
k	phase in the system
m	number of different types of atoms in the system

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Appendix A. Chemical potential at system temperature and reference pressure

The Gibbs–Helmholtz relation is given by [31]:

$$\frac{\partial}{\partial T} \left(\frac{G}{T} \right)_p = -\frac{H}{T^2} \quad \text{or} \quad \frac{\partial}{\partial T} \left(\frac{\mu_i}{T} \right)_p = -\frac{H_i}{T^2} \quad (\text{A1})$$

The chemical potential at some given temperature, T , can be calculated from Eq. (A1) by:

$$\frac{\mu_i(P_0, T)}{T} = \frac{\mu_i(P_0, T_0)}{T_0} - \int_{T_0}^T \frac{H_i(P_0, T')}{T'^2} dT' \quad (\text{A2})$$

Table C1

Thermodynamical parameters (cal/mol) for Problem 1.

	C_1	C_2	C_3	C_4	H_f	G_f
C ₆ H ₆	−8.101	1.133E−1	−7.206E−5	1.703E−8	19,819.4	30,978.3
C ₂ H ₃ N	4.8916	2.857E−2	−1.073E−5	7.650E−10	20,999.3	25,246.0
H ₂ O	7.7055	4.59E−04	2.52E−06	−8.59E−10	−57,839.39	−54,684.51

Table C2

Antoine parameters for Problem 1.

	A	B	C
C ₆ H ₆	10.8816	3823.793	−1.461
C ₂ H ₃ N	9.8653	3120.864	−37.853
H ₂ O	11.7053	3829.487	−45.622

Table C3

Thermodynamical parameters (cal/mol) for Problem 2.

	H_f	G_f
SBA	−81,883	−41,611
DSBE	−95,961	−29,876
H ₂ O	−57,839.39	−54,684.51

where $\mu_i(P_0, T_0)$ is the chemical potential at some known temperature T_0 and some pressure P_0 .

For an ideal gas, enthalpy is only a function of temperature. For a pure component, the value of $\mu_i(P, T)$ can be calculated from a known value $\mu_i(P_0, T)$ by integrating the Gibbs–Duhem equation from P_0 to P , at constant temperature. Therefore, for a pure component i :

$$\mu_i(P, T) = \left(\frac{T}{T_0} \right) \mu_i(P_0, T_0) + \int_{P_0}^P V_i(P', T) dP' - T \int_{T_0}^T \frac{H_i(T')}{T'^2} dT' \quad (\text{A3})$$

The partial molar enthalpy is calculated by [31]:

$$\left(\frac{\partial \bar{H}_i}{\partial T} \right)_p = C_{p_i} \quad (\text{A4})$$

$$C_{p_i} = C_1 + C_2 T + C_3 T^2 + C_4 T^3 \quad (\text{A5})$$

where C_1 , C_2 , C_3 and C_4 are constants.

Applying Eqs. (A4) and (A5) into Eq. (A3) and integrating over the temperature, the result is:

$$\begin{aligned} \mu_i(P, T) = & \left(\frac{T}{T_0} \right) \Delta G_{f_i}^0 + \left(1 - \frac{T}{T_0} \right) \Delta H_{f_i}^0 - C_1 \left(T \ln \frac{T}{T_0} - T + T_0 \right) \\ & - \frac{C_2}{2} (T - T_0)^2 - \frac{C_3}{6} (T^3 - 3T_0^2 T + 2T_0^3) \\ & - \frac{C_4}{12} (T^4 - 4T_0^3 T + 3T_0^4) + \int_{P_0}^P V_i(P', T) dP' \end{aligned} \quad (\text{A6})$$

Defining:

$$\begin{aligned} \mu_i^s(T) = & \left(\frac{T}{T_0} \right) \Delta G_{f_i}^0 + \left(1 - \frac{T}{T_0} \right) \Delta H_{f_i}^0 - C_1 \left(T \ln \frac{T}{T_0} - T + T_0 \right) \\ & - \frac{C_2}{2} (T - T_0)^2 - \frac{C_3}{6} (T^3 - 3T_0^2 T + 2T_0^3) - \frac{C_4}{12} (T^4 - 4T_0^3 T + 3T_0^4) \end{aligned} \quad (\text{A7})$$

The standard state is usually $P_0 = 1$ bar and $T_0 = 298.15$ K, so that:

$$\mu_i(P_0, T_0) = \Delta G_{f_i}^0 \quad (\text{A8})$$

$$H_i(T_0) = \Delta H_{f_i}^0 \quad (\text{A9})$$

Table C4

Antoine parameters for Problem 2.

	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆
SBA	51.634	−1.05E+4	0.00	−8.54E−2	6.21E−5	0.00
DSBE	106.07	−1.65E+4	0.00	−2.46E−1	2.13E−4	0.00
H ₂ O	70.435	−7.36E+3	0.00	6.95E−3	0.00	−9.00

 $\ln P = C_1 + (C_2/(C_3 + T)) + C_4 T + C_5 T^2 + C_6 \ln T$; T in K and P in atm.
Table C5

Thermodynamical parameters (cal/mol) for Problem 3.

	C ₁	C ₂	C ₃	C ₄	H _f	G _f
C ₂ H ₆	2.1529	5.11E−2	−2.004E−5	3.279E−10	−56,128.8	−40,221.6
C ₆ H ₁₂	−1.0540	0.1390	−7.449E−5	1.551E−8	−39,958.9	−39.9
H ₂ O	7.7055	4.59E−04	2.52E−06	−8.59E−10	−57,839.39	−54,684.51

Table C6

Antoine parameters for Problem 3.

	A	B	C
C ₂ H ₆	12.0588	3984.923	−39.724
C ₆ H ₁₂	9.2920	3667.705	−46.966
H ₂ O	11.7053	3829.487	−45.622

Table C7

Thermodynamical parameters (cal/mol) for Problem 4.

	C ₁	C ₂	C ₃	C ₄	H _f	G _f
C ₇ H ₈	−5.8200	1.2249E−1	−6.6085E−5	−1.1738E−8	12,029.2	29,182.6
C ₂ H ₃ N	4.8916	2.857E−2	−1.073E−5	7.650E−10	20,999.3	25,246.0
H ₂ O	7.7055	4.59E−04	2.52E−06	−8.59E−10	−57,839.39	−54,684.51

Table C8

Antoine parameters for Problem 4.

	A	B	C
C ₇ H ₈	9.5232	3171.991	−50.507
C ₂ H ₃ N	9.8653	3120.864	−37.853
H ₂ O	11.7053	3829.487	−45.622

where ΔG_{fi}^0 and ΔH_{fi}^0 are the standard molar Gibbs free energy of formation and the standard molar enthalpy of formation for component i , respectively.

Applying Eq. (A7) into Eq. (A6), the final result is:

$$\mu_i(P, T) = \mu_i^\circ(T) + \int_{P_0}^P V_i(P', T) dP' \quad (\text{A10})$$

Table C9

Thermodynamical parameters (cal/mol) for Problem 5.

	C ₁	C ₂	C ₃	C ₄	H _f	G _f
C ₁₀ H ₈	−16.432	2.0299E−1	−1.5539E−4	4.7315E−8	36,089	53,429
CO ₂	4.7323	1.75E−02	−1.33E−05	4.09E−09	−94,120.46	−94,311.66

Table C10

Parameters for Problem 5.

	T _c (K)	P _c (bar)	w
C ₁₀ H ₈	748.4	40.50	0.3020
CO ₂	304.21	73.83	0.2236

Table C11

Thermodynamical parameters (cal/mol) for Problem 6.

	C ₁	C ₂	C ₃	C ₄	H _f	G _f
Trans-2-hexen-1-ol	−2.7383	1.50E−1	−1.00E−4	2.81E−8	−48,322.2	−13,613.8
CO ₂	4.7323	1.75E−02	−1.33E−05	4.09E−09	−94,120.46	−94,311.66

Table C12

Parameters for Problem 6.

	T_c (K)	P_c (bar)	w
Trans-2-hexen-1-ol	601.76	36.73	0.724
CO ₂	304.21	73.83	0.2236

Table C13

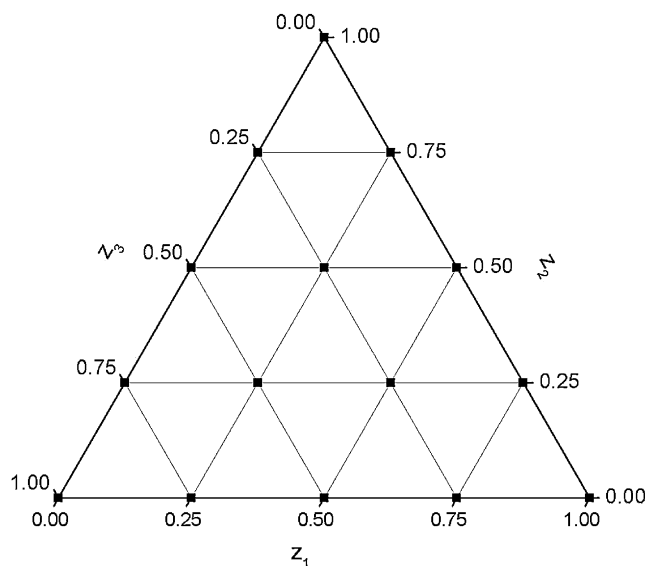
Thermodynamical parameters (cal/mol) for Problems 7 and 8.

	C_1	C_2	C_3	C_4	H_f	G_f
H ₂ O	7.7055	4.59E-04	2.52E-06	-8.59E-10	-57,839.39	-54,684.51
C ₂ H ₄	0.909	3.740E-2	-1.994E-5	4.192E-9	12,496	16,282
C ₃ H ₈ O	0.5903	7.946E-2	-4.433E-5	1.026E-8	-61,328.94	-38,695.07

Table C14

Parameters for Problems 7 and 8.

	T_c (K)	P_c (bar)	w
H ₂ O	647.3	220.5	0.34486
C ₂ H ₄	282.4	50.4	0.08625
C ₃ H ₈ O	536.8	51.7	0.62043

**Fig. B1.** Compositions vectors for $\delta = 0.25$.

Appendix B. Example of all compositions vectors generated for a ternary system

See Fig. B1.

Appendix C. Thermodynamical properties used in the case studies

See Tables C1–C14.

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